

Literature-implied answer for arXiv:2312.05617

The infinite-dihedral square has the SOS property

Status. **Literature-implied answer, full scope for the source question.** Mehta–Slofstra–Zhao ask in the introduction after Corollary 1.2 whether $\mathbb{C}Z_2^{*2} \times Z_2^{*2}$ has the SOS property; at the end of Example 6.8 they equivalently say that, for $G = Z_2^{*2}$, they do not know whether $\mathbb{C}G \times G$ is archimedean closed. The answer is yes.

Supporting theorems. Dritschel’s two-variable matrix Fejer–Riesz theorem says that every positive finite-dimensional operator-valued trigonometric polynomial in two variables is a finite sum of hermitian squares of analytic matrix polynomials [3, Introduction theorem]. Equivalently, for every n , the type-I Positivstellensatz holds for

$$M_n(\mathbb{C}[z^{\pm 1}, w^{\pm 1}])$$

with positivity understood pointwise on $\mathbb{T}^2$.

Savchuk–Schmudgen prove that if the type-I Positivstellensatz holds for $M_n(A)$, then it also holds for the finite crossed product $A \rtimes_{\alpha} F$, where F is a finite group of $*$ -automorphisms of A and $n = |F|$. Their proof embeds the crossed product into $M_n(A)$ by the regular covariant representation and applies a strong conditional expectation back to the crossed product [2, Section 5, Theorem 3].

Theorem. *Let $G = Z_2^{*2}$ be the infinite dihedral group. Then every positive element of the group algebra $\mathbb{C}[G \times G]$ is a finite sum of hermitian squares in $\mathbb{C}[G \times G]$. Hence $\mathbb{C}Z_2^{*2} \times Z_2^{*2}$ has the SOS property, or equivalently is archimedean closed.*

Proof. Write $G = \langle r, s \mid s^2 = 1, srs = r^{-1} \rangle \cong \mathbb{Z} \rtimes C_2$. Then

$$G \times G \cong \mathbb{Z}^2 \rtimes F, \quad F = C_2 \times C_2,$$

where the two generators of F invert the two coordinates of $\mathbb{Z}^2$. Consequently

$$\mathbb{C}[G \times G] \cong \mathbb{C}[z^{\pm 1}, w^{\pm 1}] \rtimes_{\alpha} F.$$

Let $A = \mathbb{C}[z^{\pm 1}, w^{\pm 1}]$. By Dritschel’s theorem, every pointwise positive element of $M_4(A)$ is a sum of hermitian squares in $M_4(A)$. Apply Savchuk–Schmudgen’s finite crossed-product transfer theorem with $|F| = 4$. It follows that every element of $A \rtimes_{\alpha} F$ that is positive in the regular covariant representation is a sum of hermitian squares inside $A \rtimes_{\alpha} F$.

It remains only to match this positivity with the representation-positive notion used by Mehta–Slofstra–Zhao. The action of the finite group F on $\mathbb{T}^2$ gives the faithful regular representation of the full crossed product $C(\mathbb{T}^2) \rtimes F$ into $M_4(C(\mathbb{T}^2))$; equivalently, the canonical conditional expectation onto $C(\mathbb{T}^2)$ is faithful. Since F is finite, these regular covariant fibres contain the finite isotropy representations at fixed points, and positivity in the faithful regular representation is exactly positivity in the full crossed-product C^* -algebra. Restricting to the algebraic crossed product gives the required representation-positive notion for $\mathbb{C}[G \times G]$. $\square$

Search evidence. Targeted searches for the exact phrases “ CZ_2^{*2} SOS property”, “archimedean closed Z_2^{*2} ”, and “infinite dihedral sums of squares group algebra” did not find a paper explicitly stating that the Mehta–Slofstra–Zhao endpoint question is resolved. The answer above is therefore recorded as an agent-identified implication of known theorems rather than as a paper whose authors explicitly advertised the resolution.

Scope. This note answers the source’s infinite-dihedral-square SOS/archimedean-closed question. It does not address the Π_2^0 -completeness conjecture for the larger free algebras, the tracial positivity conjecture, or smallest generator bounds for tracial positivity.

References

- [1] Arthur Mehta, William Slofstra, and Yuming Zhao. *Positivity is undecidable in tensor products of free algebras*. arXiv:2312.05617v2; Journal of the European Mathematical Society, online first, 2026.
- [2] Yurii Savchuk and Konrad Schmudgen. *Positivstellensätze for Algebras of Matrices*. arXiv:1004.1529v2; Linear Algebra and its Applications 436 (2012), 758–788.
- [3] Michael A. Dritschel. *Factoring non-negative operator valued trigonometric polynomials in two variables*. Mathematische Annalen 391 (2025), 515–537; published online 2024.